

# Poisson Regression (from wiki)

\* model count data

\* known as a log-linear model.

\*  $x \in \mathbb{R}^n$ .

$$\log(E(Y|X)) = \alpha + \beta x$$

if  $\theta \in \mathbb{R}^{n+1}$

$$\log(E(Y|X)) = \theta^T x \quad (x_0 = 1, \theta = (\alpha, \beta))$$

$$\Rightarrow E(Y|X) = e^{\theta^T x} = \lambda \quad (\text{for poisson distribution})$$

\*  $\theta$  can be estimated by maximum likelihood.

\* probability mass function

$$\begin{aligned} p(Y|X, \theta) &= \frac{\bar{e}^{-\lambda} \lambda^y}{y!} = \frac{\bar{e}^{-(e^{\theta^T x})} (e^{\theta^T x})^y}{y!} \\ &= \frac{e^{-e^{\theta^T x}} e^{y\theta^T x}}{y!} \end{aligned}$$

\* probability of  $m$  data points,

$$p(y_1, \dots, y_m | x_1, \dots, x_m, \theta) = \prod_{i=1}^m \frac{e^{y_i \theta^T x_i} e^{-e^{\theta^T x_i}}}{(y_i)!}$$

$L(\theta | X, Y)$  likelihood function in terms of  $\theta$

\*

$$\ell(\theta|X, Y) = \log L(\theta|X, Y)$$

$$= \sum_{i=1}^m (y_i \theta^T x_i) - e^{\theta^T x_i} - \log(y_i!)$$

$$:= \sum_{i=1}^m y_i \theta^T x_i - e^{\theta^T x_i} \quad (\because \log(y_i!) \text{ independent of } \theta)$$

\* The maximum-likelihood estimates lack a closed-form expression.  
must be found by numerical methods,  $\left( \frac{\partial \ell(\theta|X, Y)}{\partial \theta} = 0 \right)$

\* The negative log-likelihood  $[-\ell(\theta|X, Y)]$  is a convex function.

The standard convex optimization techs such as gradient can be applied to find optimal value of  $\theta$ .

\* Regularized Poisson Regression

$$\text{Max}_{\theta} \left( \sum_{i=1}^m \log \left( \frac{e^{y_i \theta^T x_i}}{y_i!} \right) - \lambda \|\theta\|_2^2 \right)$$